Collatz

Statement

Evirir the dragon has a list of n integers $a_1, a_2, \ldots, a_n$. Evirir will repeat the following procedure k times:

- For each integer a_i in the list:
  - If a_i is even, replace a_i with $\frac{a_i}{2}$.
  - Otherwise, replace a_i with $3a_i + 1$.

In the end, what is the sum of all integers in the list?

Input Format

- The first line consists of two integers n and k ($1 \leq n \leq 1000, 1 \leq k \leq 1000$).
- Then n integers follow: $a_1, a_2 \ldots, a_n$ ($1 \leq a_i \leq 10^4$).

Output Format

Output a single integer, the sum of the integers in the final list.

Sample Input 1

```
5 1
1 2 3 4 5
```

Sample Output 1

```
33
```

Explanation 1

After performing the procedure once, the list becomes $\{4, 1, 10, 2, 16\}$. Summing up, we get $4+1+10+2+16 = 33$.

Sample Input 2

```
6 3
3 1 4 1 5 9
```

Sample Output 2

```
33
```

Explanation 2

The list is changed like so: $\{3, 1, 4, 1, 5, 9\} \to \{10, 4, 2, 4, 16, 28\} \to \{5, 2, 1, 2, 8, 14\} \to \{16, 1, 4, 1, 4, 7\}$. The sum of the integers in the final list is $16 + 1 + 4 + 1 + 4 + 7 = 33$.